

Luis Sarmiento

Economist

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SUMMARY

I am an economist focusing on causal inference and energy modeling. As an experienced professional, I have developed substantial expertise in applied econometrics, integrated assessment models, and agent-based modeling. My research focuses on critical issues such as the energy transition, climate change, digitalization, environmental justice, and the study of policies to decrease environmental externalities in urban areas.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher — **Jan 2026 – Present**
MTEC Department, ETH Zurich, Zurich CH

Research Economist — **Sep 2024 – Jan 2026**
Directorate of Economic Research, Bank of Mexico, Merida MX

Environmental Economist — **Sep 2022 – Sep 2024**
Euro-Mediterranean Center on Climate Change, Milan IT

Postdoctoral Researcher — **Sep 2020 – Sep 2022**
Euro-Mediterranean Center on Climate Change, Milan IT

Research Associate — **Apr 2017 – Sep 2020**
German Institute of Economic Research, Berlin DE

Research Associate — **Nov 2019 – Aug 2020**
Hertie School of Governance, Berlin DE

Business Executive — **Sep 2012 – Sep 2014**
Banco del Bajio SA, Leon MX

EDUCATION

Doctor in Economics (*summa cum laude*) — **2017–2020**
Berlin Institute of Technology (TU Berlin), Berlin Germany

M.S. Economics and Management — **2014–2016**
Humboldt University of Berlin, Berlin Germany

B.S. International Business — **2007–2012**
University of La Salle Bajio, Leon Mexico

PUBLICATIONS

Emergency department visits and temperature: Evidence from Mexico
Sarmiento L., Colelli F.P., and Pavanello F.
Journal of Economic Behavior and Organization, 250, 107728, 2026

The air quality effects of Uber
Sarmiento L. and Kim Y.
Journal of Urban Economics, 153, 103846, 2026

◇

Occupation and temperature-related mortality in Mexico
Bressler R.D., Papp A., Sarmiento L., Shrader J., and Wilson A.
Journal of Human Resources, 2026

◇

Recycling in a Globalised Economy
Joltreau E. and Sarmiento L.
Journal of Applied Economics, 29, 2699509, 2026

◇

Policy implications of net-zero emissions: A multi-model analysis of US energy system impacts
Bistline J. et al.
Energy and Climate Change, 6, 100191, 2025

◇

Residential Solar Adoption: Main Determinants and the Future of Small-Scale Photovoltaics
Reissl S., Sarmiento L., and Emmerling J.
Environmental Research - Energy, 2(3), 035002, 2025

◇

Health and air pollutant impacts of Net Zero CO2 by 2050 from the Energy Modeling Forum 37
Loughlin D., Sarmiento L., Emmerling J., et al.
Energy and Climate Change, 5, 100165, 2024

◇

Carbon Management Technology Pathways for Reaching a US Economy-Wide Net-Zero Emissions Goal
Binsted M. et al.
Energy and Climate Change, 5, 100154, 2024

◇

Comparing Net Zero pathways across the Atlantic: Inter-comparison between EMF-37 and ECEMF
Sarmiento L. et al.
Energy and Climate Change, 5, 100144, 2024

◇

Equity Implications of Net-Zero Emissions: Energy Expenditures across Income Classes
Bistline J. et al.
Energy and Climate Change, 5, 100118, 2024

◇

Policy Reversals in Transitional Markets: Changing Marginal Cost to Physical Order Dispatch
Gutierrez-Meave R., Rosellon J., and Sarmiento L.
Economics of Energy and Environmental Policy, 13(1), 57-82, 2024

◇

The Air Quality and Well-Being Effects of Low Emission Zones
Sarmiento L., Wagner N., and Zaklan A.
Journal of Public Economics, 227, 105014, 2023

Air Pollution and Court Decisions: Evidence from Ten Million Penal Cases in India
Sarmiento L. and Nowakowski A.
Environmental and Resource Economics, 86, 605-644, 2023

◇

The Nonlinear Effects of Pollution on Crime: Evidence from Mexico City and New York
Sarmiento L.
Environmental Research - Health, 1(2), 021001, 2023

◇

Congestion, Air Quality, and Space
Tassinari F. and Sarmiento L.
Handbook of Labor, Human Resources and Population Economics, 1-18, 2023

◇

Air Pollution and the Productivity of High-Skill Labor: Evidence from Court Hearings
Sarmiento L.
Scandinavian Journal of Economics, 124, 301-332, 2022

◇

Mexico and US Power Systems Under Variations in Natural Gas Prices
Sarmiento L. et al.
Energy Policy, 156, 112378, 2021

◇

The Role of Energy Storage in the Uptake of Renewable Energy: A Model Comparison Approach
Giarola S. et al.
Energy Policy, 151, 112159, 2021

◇

The Effects of the 2013 Floods on Germany's Road Freight Traffic
Gabela J.G.F. and Sarmiento L.
Transportation Research Part D, 82, 102274, 2020

◇

Analysing Scenarios for Renewable Energy Integration in the Mexican Energy System
Sarmiento L. et al.
Energies, 12(17), 3270, 2019

UNDER REVIEW

Income Shocks, Adaptation, and Temperature-Related Mortality: Evidence from the Mexican Labor Market
Sarmiento L., Gilli M., Pavanello F., and Shayegh S.
Journal of the Association of Environmental and Resource Economists (Reject and Resubmit)

◇

When the Boundary Layer Drops: Air Quality and Healthcare Use in Mexico
Basaglia P. and Sarmiento L.
Journal of Environmental Economics and Management

◇

The Unintended Consequences of Low Emission Zones: Evidence from Madrid Central
Santonja A., Sarmiento L., and Wagner N.
Journal of Environmental Economics and Management (Revise and Resubmit)

◇

The Air Quality, Health, and Labor Supply Effects of Crop-Residue Burning
Sarmiento L. and Arellano J.
American Journal of Agricultural Economics

WORKING PAPERS

The Convexity of Hurricane Damages
Cohen F., Sarmiento L., and Stuka Y.

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Racial Inequalities in Regional Air Pollution Exposure in Mexico
Sarmiento L. and Gutierrez-Meave R.

◇

The Transnational Effect of Temperature on Remittances
Ares de Parga-Regalado G., Osuna-Gomez D., and Sarmiento L.

◇

Inflation Smoothing Through Natural Disaster Relief Funds
Arellano J., Llamosas I., McLiberty I., Sarmiento L., and Solorzano D.

◇

Temperatures and Distribution Blackouts: From Mechanisms to Health Impacts
Colelli F.P., Pavanello F., and Sarmiento L.

◇

Indoor Air Quality and Student Welfare: The Effect of Air Purifiers in Schools
Bonan J., Granella F., Renna S., and Sarmiento L.

◇

The Global Economic Burden of Air Pollution
Basaglia P., Drupp M.A., Sager L., Sarmiento L., and Schaumann F.

◇

Rural Development and Environmental Trade-offs in Mexico's Flagship Agroforestry Programme
Meier S. and Sarmiento L.

◇

Migrant Networks as Climate Insurance
Ares de Parga-Regalado G., Osuna-Gomez D., and Sarmiento L.

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Low Emission Zones and Fossil Fleet Renewal in Spain
Sarmiento L. and Amich M.

GRANTS AND FUNDING

- **Evaluating Non-price and Economic Instruments for Demand Evening in Mobility and Transport (EVEN-DEMAND)**, Swiss Federal Office of Energy (SFOE), Principal Investigator, CHF 220,000, 2027-2029

HONORS AND AWARDS

- **Honorable Mention, Premio Banamex de Economía**, Research category, 2026. *The Transnational Effect of Temperature on Remittances*, with D. Osuna-Gómez and G. Ares de Parga-Regalado

TEACHING

Courses Taught

- **Designing Public Policy Research (Quasi-Experimental Methods)**, ETH Zurich, Fall 2026
- **Environmental Economics and Policy**, University of Bologna, Spring 2026
- **Environmental Economics and Policy**, University of Bologna, Spring 2025
- **Environmental Economics and Policy**, University of Bologna, Winter 2024
- **Global Environmental Challenges**, Politecnico Di Milano, Spring 2024
- **Environmental Economics and Policy**, University of Bologna, Spring 2024
- **Environmental Economics and Policy**, University of Bologna, Spring 2023
- **Applied Econometrics for Energy and Env. Economics**, Berlin University of Technology, Summer 2020
- **Applied Econometrics for Energy and Env. Economics**, Berlin University of Technology, Summer 2019

Visiting Scholar

- **Comparing decarbonisation scenarios across the Atlantic**, Univ. of Tokyo, Fall 2024
- **Environmental Economics and Policy**, KDI School of Public Policy, Fall 2024
- **Environmental Economics and Policy**, KDI School of Public Policy, Spring 2023
- **Environmental Economics**, University of Verona, Spring 2023

TECHNICAL SKILLS & LANGUAGES

Software: R (RStudio), Python, Quarto, Git, GAMS, Julia

Methods: Difference-in-Differences, Regression Discontinuity, Instrumental Variables, Panel Data, Machine Learning, Agent-Based Modeling, Integrated Assessment Models

Languages: Spanish (native), English (fluent), Italian (advanced), German (intermediate)

SELECTED CONFERENCES

- **World Congress of Environmental and Resource Economists**, Lisbon PT, Summer 2026
- **Gothenburg Environmental Economics Workshop**, Gothenburg SE, Summer 2026
- **Mannheim Energy Conference**, Mannheim DE, Spring 2026
- **Essen Health Conference**, Essen DE, Spring 2026
- **First Colloquium of Regional Economists**, Merida MX, Fall 2025
- **BIS Conference on Climate and the Environment**, Bogota CO, Fall 2025
- **First Latin-American Environmental Economics Conference**, Bogota CO, Fall 2025
- **Mapping Environmental and Energy Policies Workshop**, Frankfurt DE, Spring 2025
- **EAERE Annual Conference**, Leuven BE, Summer 2024
- **MEEP Workshop**, Hamburg DE, Summer 2024
- **IAERE Annual Conference**, Pescara IT, Spring 2024
- **IEA World Congress**, Medellin CO, Fall 2023
- **ICEE World Conference**, Cagliari IT, Spring 2023
- **IAERE Annual Conference**, Napoli IT, Winter 2023
- **EAERE Annual Conference**, Rimini IT, Summer 2022

REFERENCES

Christian von Hirschhausen

Professor, TU Berlin

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Moritz Drupp

Associate Professor, ETH Zurich

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